COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

STAT 5703

Statistical Inference and Modeling

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10. Multivariate Normal Distribution

Univariate Normal (Gaussian) Distribution

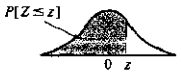
- Bell-shaped distribution with tendency for individuals to clump around the group median/mean
- Used to model many biological phenomena
- Many *estimators* have approximately normal sampling distributions (Central Limit Theorem)
- Notation: $X \sim \mathrm{N}(\mu, \sigma)$ where μ is mean and σ^2 is the variance

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}}, -\infty < x < \infty, -\infty < \mu < \infty, \sigma > 0$$

Note: To find probabilities about $X \sim \mathrm{N}(\mu, \sigma)$ on the exams you must convert it to $Z = \frac{X-\mu}{\sigma} \sim N(0,1)$ and then use the z-table.

Second Decimal Place of z

TABLE 1 STANDARD NORMAL PROBABILITIES



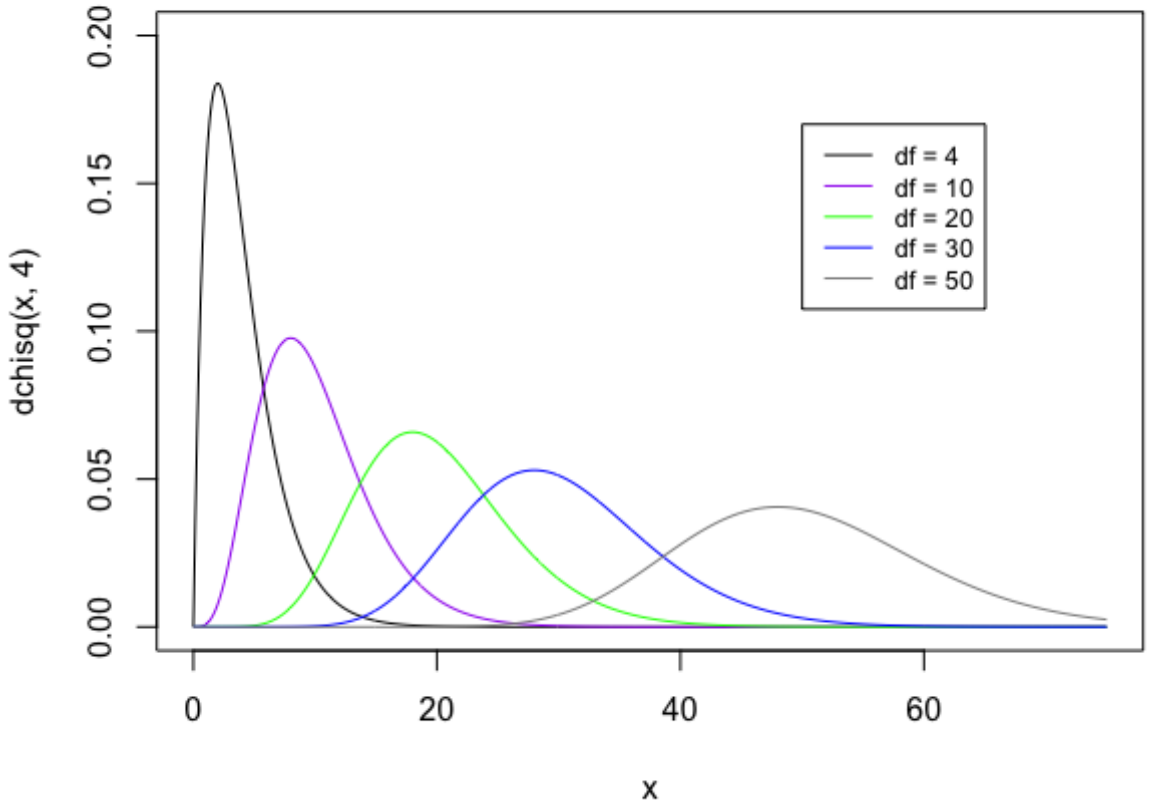
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7703	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998
.5	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998

Integer part and first decimal place of z

Chi-Square Distribution

- If $Z_1, \ldots, Z_\nu$ are independent N(0, 1) variables, then
$$X = Z_1^2 + \cdots + Z_\nu^2$$
is called a chi-square distribution with ν degrees of freedom.
- Notation: $X \sim \chi_\nu^2$, where ν are the degrees of freedom
- The pdf is
$$f(x) = \frac{1}{\Gamma\left(\frac{\nu}{2}\right) 2^{\frac{\nu}{2}}} x^{\frac{\nu}{2}-1} e^{-\frac{x}{2}}, x > 0, \nu > 0$$
- E(X) = ν, Var(X) = 2 ν
- Special case: $Z \sim N(0,1) \Rightarrow Z^2 \sim \chi_1{}^2$
- If $X_1, \ldots, X_n$ are independent chi-square variables, and $X_i \sim \chi_{\nu_i}^2$
$$\sum_{i=1}^{n} X_i \sim \chi_{\sum_{i=1}^{n} \nu_i}^2$$

Chi-Square Distributions



Note: Chi-square table will be provided to the exam, if it is needed

Student's *t*-Distribution

- Indexed by "degrees of freedom ν" $X \sim t_\nu$
- $Z \sim N(0,1)$, $X \sim \chi_\nu^2$
- Assuming Independence of Z and X:

$$T = \frac{Z}{\sqrt{X/\nu}} \sim t_\nu$$

Density Function:

$$f(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\nu}\,\Gamma\left(\frac{\nu}{2}\right)}\left(1+\frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}} \quad -\infty < t < \infty \quad \nu > 0 \qquad E\{T\} = 0\ (\nu > 1) \quad V\{T\} = \frac{\nu}{\nu-2}\ (\nu > 2)$$

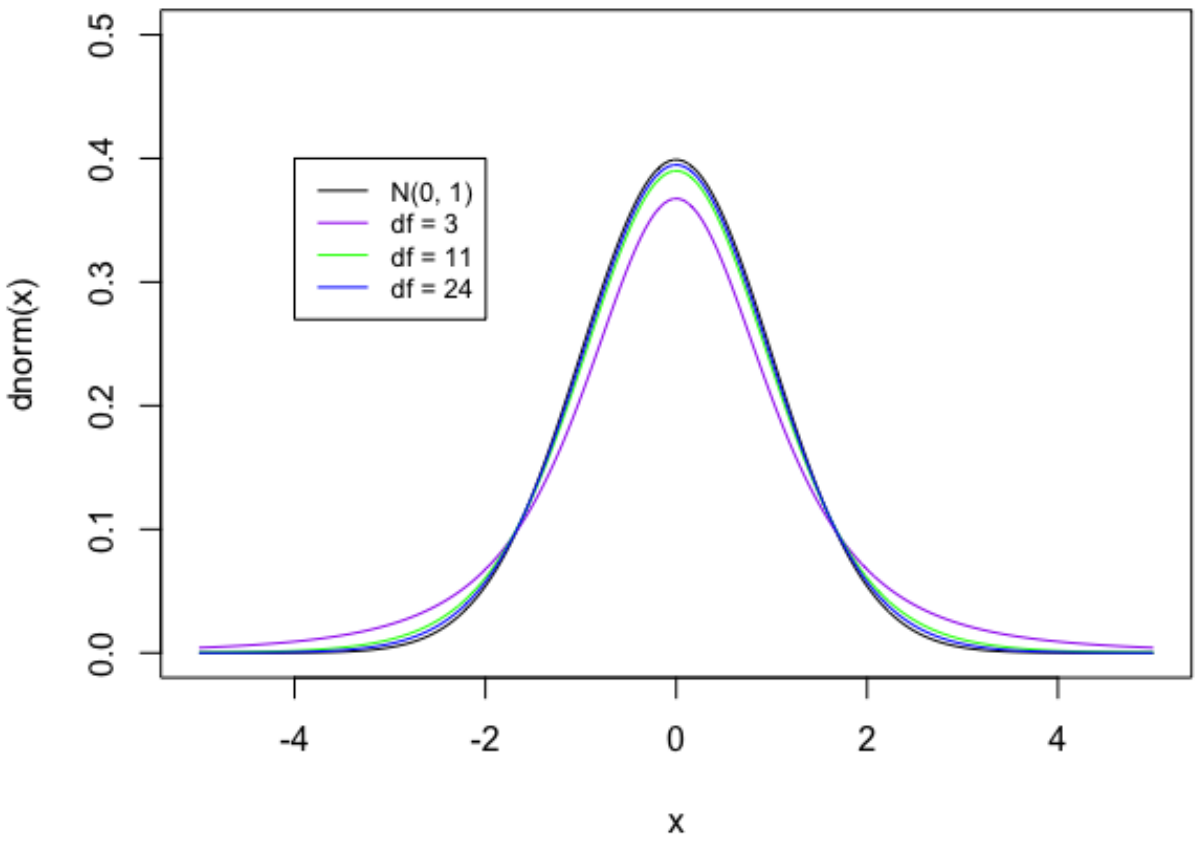
Obtaining Probabilities/Quantiles in Python:

To obtain: $F(t) = P(T \le t)$: scipy.stats.t.cdf(t, ν)

p^{th} quantile: scipy.stats.t.ppf(p, ν)

Tables usually give upper tail cut-off values for some upper probabilities

t distributions



Critical Values for Student's t-Distribution

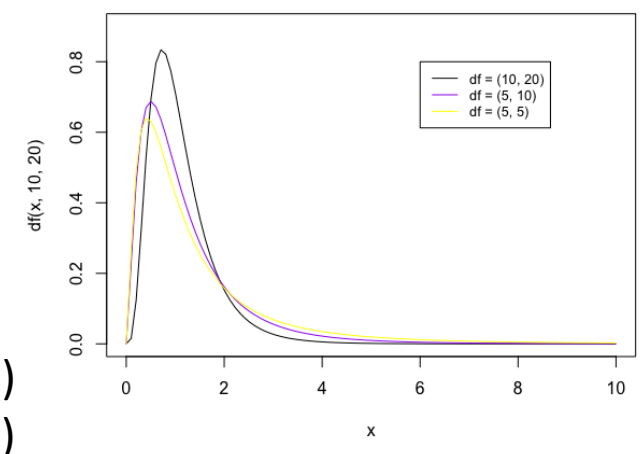
t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	**0.50**	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**	**0.0005**
two-tails	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**	**0.01**	**0.002**	**0.001**
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
					Confidence Level						

t-table.xls 7/14/2007

F - Distribution

- Arises often as a test statistic in ANOVA and regression.
- Indexed by two degrees of freedom $\nu_1 > 0$, $\nu_2 > 0$
- Notation: $W \sim F_{n_1, n_2}$
- Definition: assume independent chi square $X_1 \sim \chi^2_{n_1}$, $X_2 \sim \chi^2_{n_2}$
- Then $W = \frac{X_1/\nu_1}{X_2/\nu_2} \sim F_{\nu_1,\nu_2}$
- pdf: $f(w) = \frac{\Gamma\left(\frac{\nu_1+\nu_2}{2}\right)}{\Gamma\left(\frac{\nu_1}{2}\right)\Gamma\left(\frac{\nu_2}{2}\right)} \left(\frac{\nu_1}{\nu_2}\right)^{\frac{\nu_1}{2}} w^{\frac{\nu_1}{2}-1} \left(\frac{1}{2}\right)^{-\frac{\nu_1+\nu_2}{2}} \left(1 + \frac{w\nu_1}{\nu_2}\right)^{-\frac{\nu_1+\nu_2}{2}}, w > 0$
- Mean and variance:
- $E(W) = \frac{\nu_2}{\nu_2-2}$, for $\nu_2 > 2$
- $\mathrm{Var}(W) = \left(\frac{\nu_2}{\nu_2-2}\right)^2 \frac{\nu_1+\nu_2-2}{\nu_1(\nu_2-4)}$, for $\nu_2 > 4$



Obtaining Probabilities/Quantiles in Python:

To obtain: $F(w) = P(W \le w)$: scipy.stats.f.cdf(w, ν_1, ν_2)

p^{th} quantile: scipy.stats.f.ppf(p, ν_1, ν_2)

Tables usually give upper tail cut-off values for some upper probabilities

Critical Values for F-distributions

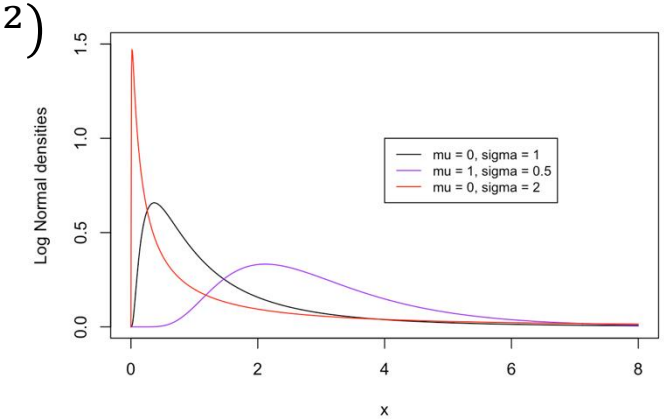
P(F ≤ Table Value) = 0.95

F-table of Critical Values of α = 0.05 for F(df1, df2)

	DF1=1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
DF2=1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.95	248.01	249.05	250.10	251.14	252.20	253.25	254.31
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.37
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

Log-normal Distribution

- If $Z \sim N(0,1), \mu \in \mathbb{R}, \sigma > 0$, then $X = e^{\mu + \sigma Z}$ is called *log-normal* distribution
- The pdf is $f(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{\sigma\sqrt{2}}}, x > 0$
- One of the most common applications where log-normal distributions are used in finance is in the analysis of stock prices and income distributions.
- The mean is $e^{\mu + \frac{1}{2}\sigma^2}$
- The second moment is $\mu_2 = e^{2(\mu + \sigma^2)}$
- The median is e^{μ}



Obtaining Probabilities/Quantiles in R:

Density $f(x)$:	dlnorm(x, μ, σ)
To obtain: $F(x) = P(X \le x)$:	plnorm(x, μ, σ)
p^{th} quantile:	qlnorm(x, μ, σ)

Bivariate Normal Distribution

If the random vector (X_1, X_2) has a pdf

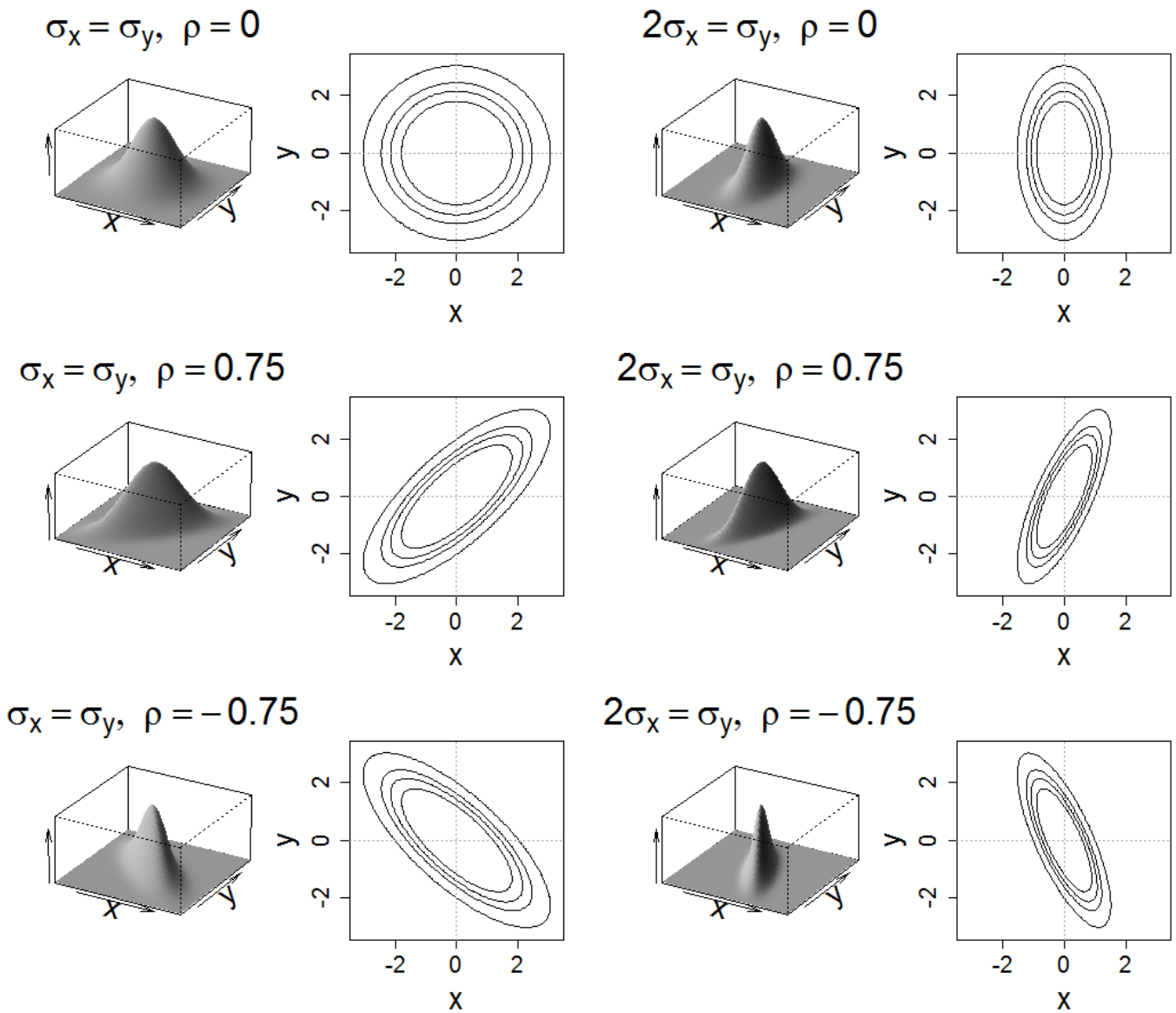
$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-½\, Q(x_1,x_2)}$$

where

$$Q(x_1, x_2) = \frac{\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x_1-\mu_1}{\sigma_1}\right)\left(\frac{x_2-\mu_2}{\sigma_2}\right) + \left(\frac{x_2-\mu_2}{\sigma_2}\right)^2}{1-\rho^2}$$

Then the distribution is called the **bivariate Normal distribution.**

The parameters are μ_1, μ_2 , σ_1, σ_2 and ρ.


$\sigma_x = \sigma_y,\ \rho = 0$
$2\sigma_x = \sigma_y,\ \rho = 0$
$\sigma_x = \sigma_y,\ \rho = 0.75$
$2\sigma_x = \sigma_y,\ \rho = 0.75$
$\sigma_x = \sigma_y,\ \rho = -0.75$
$2\sigma_x = \sigma_y,\ \rho = -0.75$
x
y
-2
0
2

Contour Plots

The bivariate normal pdf

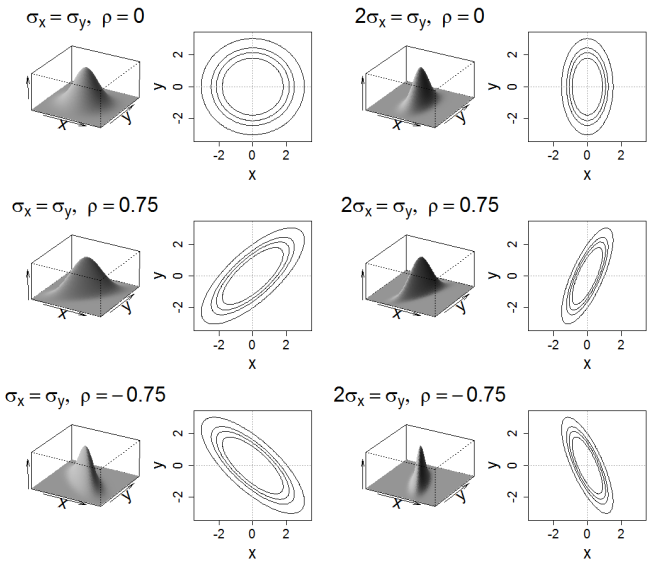
$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-½\, Q(x_1,x_2)}$$

where

$$Q(x_1, x_2) = \frac{\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x_1-\mu_1}{\sigma_1}\right)\left(\frac{x_2-\mu_2}{\sigma_2}\right) + \left(\frac{x_2-\mu_2}{\sigma_2}\right)^2}{1-\rho^2}$$

It can be shown that $f(x_1, x_2) = c$ is constant.

iff x_1, x_2 lie on an ellipse centered at μ_1, μ_2 .



Q: Why do we care about this?

A: It is related to confidence regions

(see later slides)

Marginal and Conditional Distributions

Marginal distributions for the Bivariate Normal distribution

It can be shown that in the case of (X_1, X_2) having bivariate normal distribution, then the marginal distribution of X_i is Normal with mean μ_i and standard deviation $\sigma_{i,}$ $i = 1, 2$

Thus, the marginal distribution of x_2 is Normal with mean μ_2 and standard deviation σ_2.

Similarly, the marginal distribution of x_1 is Normal with mean μ_1 and standard deviation σ_1.

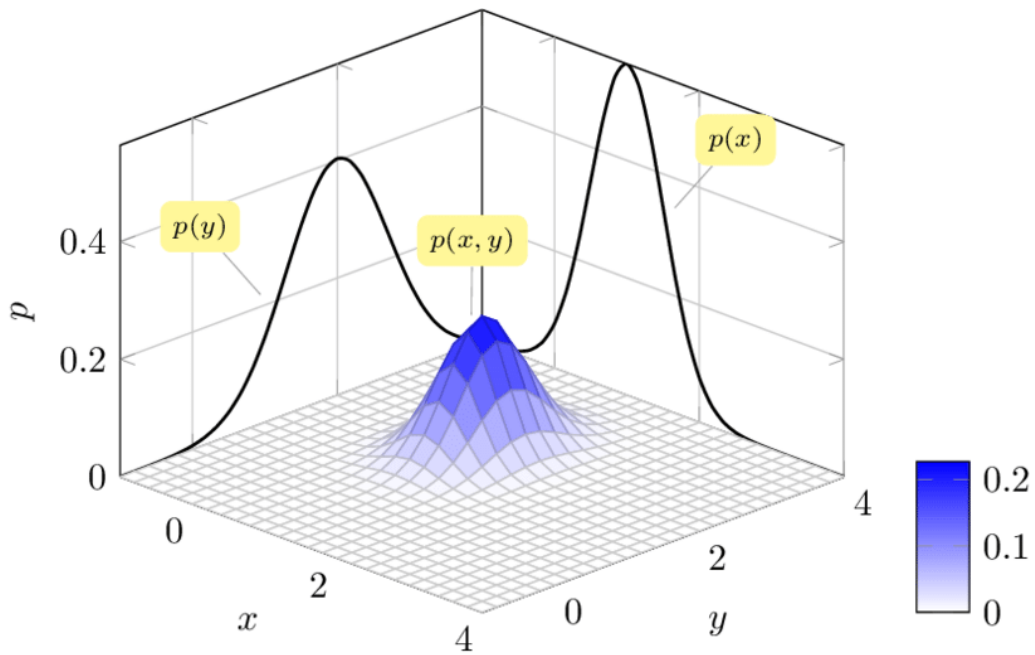
Conditional Distributions of the Bivariate Normal Distribution

If (X_1, X_2) follows the bivariate normal distribution, it can be shown that the conditional distributions of X_i given X_j , $i, j = 1, 2$ is normal with:

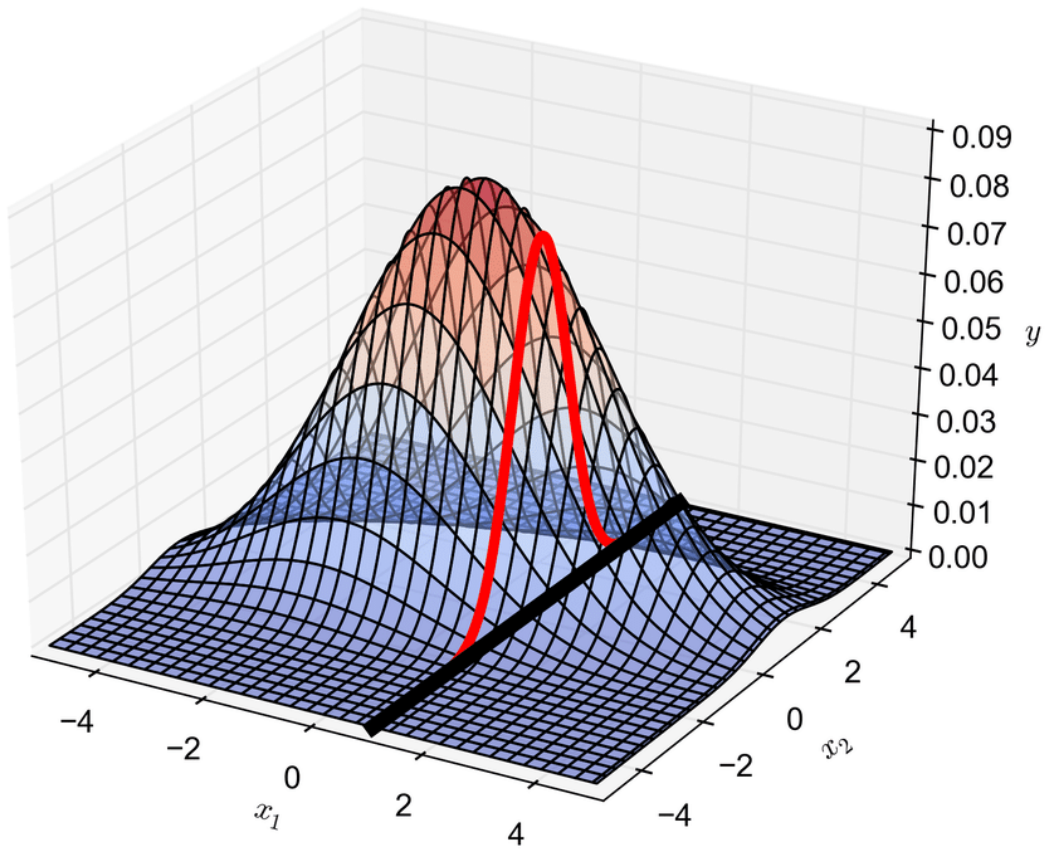
mean $$\boxed{\mu_{i|j} = \mu_i + \rho \frac{\sigma_i}{\sigma_j}\left(x_j - \mu_j\right)}$$

standard deviation $$\boxed{\sigma_{i|j} = \sigma_i \sqrt{1-\rho^2}}$$

Bivariate normal distribution with marginal distributions



Bivariate normal distribution with conditional distribution



https://demonstrations.wolfram.com/TheBivariateNormalAndConditionalDistributions/

Multivariate Normal Distribution

Notation: $\boldsymbol{X} \sim \mathrm{N}_p(\boldsymbol{\mu_X}, \boldsymbol{\Sigma_X})$, where:

$$\boldsymbol{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{bmatrix}, \boldsymbol{\mu_X} = E(\boldsymbol{X}) = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{bmatrix}, \boldsymbol{\Sigma_X} = \sigma^2(\boldsymbol{X}) = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1p} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1p} & \sigma_{2p} & \cdots & \sigma_{pp} \end{bmatrix}$$

Multivariate normal density function:

$$f(\boldsymbol{x}) = (2\pi)^{-p/2} \left| \boldsymbol{\Sigma_X}^{-1/2} \right| e^{-\frac{1}{2}(\boldsymbol{X}-\boldsymbol{\mu_X})'\boldsymbol{\Sigma}_{\boldsymbol{X}}^{-1}(\boldsymbol{X}-\boldsymbol{\mu_X})}$$

Results:

$$X_i \sim N(\mu_i, \sigma_{ii}), \qquad i = 1, \dots, p$$
$$\mathrm{cov}\left(X_i, X_j\right) = \sigma_{ij}, \qquad i \neq j$$

Multivariate Normal – Conditional Distributions

Let $\boldsymbol{X}_1$ be $q \times 1$ and $\boldsymbol{X}_2$ be $(p - q) \times 1$ vectors, and

$$\boldsymbol{X} = \begin{bmatrix} \boldsymbol{X}_1 \\ \boldsymbol{X}_2 \end{bmatrix} \sim \mathrm{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \qquad \boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \qquad \boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{bmatrix}$$

Then $\boldsymbol{X}_1 \mid \boldsymbol{X}_2 = \boldsymbol{x}_2 \sim \mathrm{N}_q\,(\boldsymbol{\mu}_{1|2}, \boldsymbol{\Sigma}_{1|2})$ with
mean $\boldsymbol{\mu}_{1|2} = \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\boldsymbol{x}_2 - \boldsymbol{\mu}_2)$ and covariance matrix

$$\boldsymbol{\Sigma}_{1|2} = \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$$

Note: the conditional mean depends on the specific value $\boldsymbol{x}_2$ but the covariance matrix does not.

Special case: $p = 2$, $q = 1$.
Then $X_1 \mid X_2 = x_2 \sim \mathrm{N}(\mu_{1|x2}, \sigma_{1|x2})$
where

$$\mu_{1|x_2} = \mu_1 + \frac{\sigma_{12}}{\sigma_{22}}(x_2 - \mu_2), \sigma_{1|x_2} = \sigma_{11} - \frac{\sigma_{12}^2}{\sigma_{22}} = \sigma_{11}(1 - \rho_{12}^2)$$

This result is the basis of linear regression models!

Summary when p = 2

Joint Distribution:

$$f(x_1,x_2)=\frac{1}{2\pi\sqrt{\sigma_1^2\sigma_2^2\left(1-\rho^2\right)}}\exp\left\{-\left(\frac{1}{2\left(1-\rho^2\right)}\right)\left[\frac{\left(x_1-\mu_1\right)^2}{\sigma_1^2}-\frac{2\rho\left(x_1-\mu_1\right)\left(x_2-\mu_2\right)}{\sigma_1\sigma_2}+\frac{\left(x_2-\mu_2\right)^2}{\sigma_2^2}\right]\right\} \quad -\infty<x_1,x_2<\infty$$

Marginal (aka Unconditional) Distributions:

$$f_1\left(x_1\right)=\frac{1}{\sqrt{2\pi\sigma_1^2}}\exp\left\{-\frac{\left(x_1-\mu_1\right)^2}{2\sigma_1^2}\right\} \quad -\infty<x_1<\infty$$

$$f_2\left(x_2\right)=\frac{1}{\sqrt{2\pi\sigma_2^2}}\exp\left\{-\frac{\left(x_2-\mu_2\right)^2}{2\sigma_2^2}\right\} \quad -\infty<x_2<\infty$$

$$X_1\sim N\left(\mu_1,\sigma_1^2\right) \qquad X_2\sim N\left(\mu_2,\sigma_2^2\right)$$

Conditional Distributions:

$$f\left(x_2\mid x_1\right)=\frac{1}{\sqrt{2\pi\sigma_2^2\left(1-\rho^2\right)}}\exp\left\{-\left(\frac{1}{2\left(1-\rho^2\right)\sigma_2^2}\right)\left[x_2-\left(\mu_2+\frac{\left(x_1-\mu_1\right)\rho\sigma_2}{\sigma_1}\right)\right]^2\right\} \quad -\infty<x_2<\infty$$

$$f\left(x_1\mid x_2\right)=\frac{1}{\sqrt{2\pi\sigma_1^2\left(1-\rho^2\right)}}\exp\left\{-\left(\frac{1}{2\left(1-\rho^2\right)\sigma_1^2}\right)\left[x_1-\left(\mu_1+\frac{\left(x_2-\mu_2\right)\rho\sigma_1}{\sigma_2}\right)\right]^2\right\} \quad -\infty<x_1<\infty$$

$$X_2\mid X_1=x_1\sim N\left[\mu_2+\frac{\left(x_1-\mu_1\right)\rho\sigma_2}{\sigma_1},\sigma_2^2\left(1-\rho^2\right)\right] \qquad X_1\mid X_2=x_2\sim N\left[\mu_1+\frac{\left(x_2-\mu_2\right)\rho\sigma_1}{\sigma_2},\sigma_1^2\left(1-\rho^2\right)\right]$$

Multivariate Normal Properties

- If $\boldsymbol{X} = (X_1, \ldots, X_p) \sim \mathrm{N}_p(\boldsymbol{\mu_x}, \boldsymbol{\Sigma_x})$, then for $1 \leq k \neq l \leq p$,
  $\mathrm{cov}(X_k, X_l) = 0$ iff X_k and X_l are independent

- If $a \in \mathbb{R}^p$ and $\boldsymbol{A}$ is an $r \times p$ matrix of constants, where $\mathrm{rank}(\boldsymbol{A}) = r \leq p$, then:

$$\boldsymbol{W} = \boldsymbol{a} + \boldsymbol{A}\boldsymbol{X} \sim \mathrm{N}_r(\boldsymbol{a} + \boldsymbol{A}\boldsymbol{\mu_x}, \boldsymbol{A}\boldsymbol{\Sigma_x}\boldsymbol{A}')$$

- Let $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$ be independent with a common covariance matrix:

$$\boldsymbol{X}_i \sim \mathrm{N}_p(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}), \; i = 1, \ldots, n$$

Define

$$\boldsymbol{W} = a_1\boldsymbol{X}_1 + \ldots + a_n\boldsymbol{X}_n$$

Then

$$\boldsymbol{W} \sim N_p\left(\sum_{i=1}^{n} a_i \boldsymbol{\mu}_i, \left(\sum_{i=1}^{n} a_i^2\right)\boldsymbol{\Sigma}\right)$$

Results Involving Multivariate Normal

Theorem: Let $\boldsymbol{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Then:

(a) $(\boldsymbol{X} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1}(\boldsymbol{X} - \boldsymbol{\mu}) \sim \chi^2_p$

(b) The probability that $\boldsymbol{X}$ is inside the solid ellipsoid

$$\{\boldsymbol{X}: (\boldsymbol{X} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1}(\boldsymbol{X} - \boldsymbol{\mu}) \leq \chi^2_p(\alpha)\}$$

is $1 - \alpha$, where $\chi^2_p(\alpha)$ denotes the upper α percentile of the χ^2_p distribution.

Univariate case: $X \sim N_1(\mu, \sigma^2)$

$$Z = \frac{X - \mu}{\sigma} \sim N(0,1)$$

$$(x-\mu)\sigma^{-2}(x-\mu) = Z^2 \sim \chi^2_1$$

Example

Let $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_{60}$ be a random sample of size 60 from a four-variate normal distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. Specify the distribution of:

(a) $\overline{\boldsymbol{X}}$

(b) $(\boldsymbol{X}_1 - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1}(\boldsymbol{X}_1 - \boldsymbol{\mu})$

(c) $n(\overline{\boldsymbol{X}} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1}(\overline{\boldsymbol{X}} - \boldsymbol{\mu})$

Example - Solution

Let $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_{60}$ be a random sample of size 60 from a four-variate normal distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. Specify the distribution of:

(a) $\bar{\boldsymbol{X}}$

(b) $(\boldsymbol{X}_1 - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1}(\boldsymbol{X}_1 - \boldsymbol{\mu})$

(c) $n(\bar{\boldsymbol{X}} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1}(\bar{\boldsymbol{X}} - \boldsymbol{\mu})$

Solution:

Given: $n = 60$, $p = 4$ and $X_i \sim \mathrm{N}_4(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $i = 1, \ldots, 60$

a) $\bar{X}$ is a 4×1 vector:

$$\bar{\boldsymbol{X}} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{1}{n}x_1 + \frac{1}{n}x_2 + \cdots + \frac{1}{n}x_n$$

Apply this result: $\boldsymbol{W} = \boldsymbol{a} + \boldsymbol{A}\boldsymbol{X} \sim \mathrm{N}_r(\boldsymbol{a} + \boldsymbol{A}\boldsymbol{\mu}_{\boldsymbol{x}}, \boldsymbol{A}\boldsymbol{\Sigma}_{\boldsymbol{x}}\boldsymbol{A}')$ with $\boldsymbol{a} = 0$

$$\boldsymbol{A} = \left(\frac{1}{n}, \ldots, \frac{1}{n}\right), \boldsymbol{\mu}_{\boldsymbol{x}} = (\mu, \ldots, \mu)'$$

$$\Rightarrow \bar{\boldsymbol{X}} \sim N_4\left(\frac{1}{n}\boldsymbol{\mu} + \frac{1}{n}\boldsymbol{\mu} + \ \ldots + \frac{1}{n}\boldsymbol{\mu}, \frac{1}{n^2}\boldsymbol{\Sigma} + \ \ldots + \frac{1}{n^2}\boldsymbol{\Sigma}\right) = \mathrm{N}_4\left(\boldsymbol{\mu}, \frac{1}{n}\boldsymbol{\Sigma}\right)$$

Example - Solution

Let $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_{60}$ be a random sample of size 60 from a four-variate normal distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. Specify the distribution of:

(a) $\bar{\boldsymbol{X}}$

(b) $(\boldsymbol{X}_1 - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1}(\boldsymbol{X}_1 - \boldsymbol{\mu})$

(c) $n(\bar{\boldsymbol{X}} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1}(\bar{\boldsymbol{X}} - \boldsymbol{\mu})$

Solution:

Given: $n = 60$, $p = 4$ and $X_i \sim \mathrm{N}_4(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $i = 1, \ldots, 60$

b) Apply the result that $(\boldsymbol{X} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1}(\boldsymbol{X} - \boldsymbol{\mu}) \sim \chi^2_p$

$(\boldsymbol{X}_1 - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1}(\boldsymbol{X}_1 - \boldsymbol{\mu}) \sim \chi^2_4$

c) Exercise.

Multivariate Normal Likelihood Function

Let $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Then the joint pdf is:

$$\prod_{i=1}^{n}\left[(2\pi)^{-\frac{p}{2}}|\Sigma|^{-\frac{1}{2}}e^{-\frac{1}{2}(x_i-\boldsymbol{\mu})'\Sigma^{-1}(\boldsymbol{x}_i-\boldsymbol{\mu})}\right] = (2\pi)^{-\frac{np}{2}}|\Sigma|^{-\frac{n}{2}}e^{-\frac{1}{2}\sum_{i=1}^{n}(\boldsymbol{x}_i-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_i-\boldsymbol{\mu})}$$

Now, working with the exponential term multiplied by -1/2, we have:

$$\sum_{i=1}^{n}(\boldsymbol{x}_i-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_i-\boldsymbol{\mu}) = \mathrm{tr}\left[\sum_{i=1}^{n}(\boldsymbol{x}_i-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_i-\boldsymbol{\mu})\right]$$

$$= \mathrm{tr}\left[\sum_{i=1}^{n}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_i-\boldsymbol{\mu})(\boldsymbol{x}_i-\boldsymbol{\mu})'\right]$$

Note: the first equality is because $c = \mathrm{tr}(c)$ for any scalar c.
The second equality follows from $\mathrm{tr}(\boldsymbol{AB}) = \mathrm{tr}(\boldsymbol{BA})$, where $\boldsymbol{A} = (\boldsymbol{x}_i-\boldsymbol{\mu})'$, $\boldsymbol{B} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_i-\boldsymbol{\mu})$

Multivariate Normal Likelihood Function

Recall the univariate result: $\sum_{i=1}^{n}(x_i-\mu)^2 = \sum_{i=1}^{n}(x_i-\bar{x})^2 + \sum_{i=1}^{n}(\bar{x}-\mu)^2$

Similar identity holds in the multivariate case:

$$\sum_{i=1}^{n}(\boldsymbol{x}_i-\boldsymbol{\mu})(\boldsymbol{x}_i-\boldsymbol{\mu})' = \sum_{i=1}^{n}(\boldsymbol{x}_i-\overline{\boldsymbol{x}}+\overline{\boldsymbol{x}}-\boldsymbol{\mu})(\boldsymbol{x}_i-\overline{\boldsymbol{x}}+\overline{\boldsymbol{x}}-\boldsymbol{\mu})'$$

$$= \sum_{i=1}^{n}(\boldsymbol{x}_i-\overline{\boldsymbol{x}})(\boldsymbol{x}_i-\overline{\boldsymbol{x}})' + \sum_{i=1}^{n}(\overline{\boldsymbol{x}}-\boldsymbol{\mu})(\overline{\boldsymbol{x}}-\boldsymbol{\mu})' + 2\sum_{i=1}^{n}(\boldsymbol{x}_i-\overline{\boldsymbol{x}}))(\overline{\boldsymbol{x}}-\boldsymbol{\mu})'$$

$$= \sum_{i=1}^{n}(\boldsymbol{x}_i-\overline{\boldsymbol{x}})(\boldsymbol{x}_i-\overline{\boldsymbol{x}})' + n(\overline{\boldsymbol{x}}-\boldsymbol{\mu})(\overline{\boldsymbol{x}}-\boldsymbol{\mu})'$$

Therefore, the joint likelihood function is:

$$(2\pi)^{-\frac{np}{2}}|\Sigma|^{-\frac{n}{2}}e^{-\frac{1}{2}\sum_{i=1}^{n}(\boldsymbol{x}_i-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_i-\boldsymbol{\mu})} = (2\pi)^{-\frac{np}{2}}|\Sigma|^{-\frac{n}{2}}e^{-\frac{1}{2}\mathrm{tr}\left[\sum_{i=1}^{n}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_i-\boldsymbol{\mu})(\boldsymbol{x}_i-\boldsymbol{\mu})'\right]}$$

$$= (2\pi)^{-\frac{np}{2}}|\Sigma|^{-\frac{n}{2}}e^{-\frac{1}{2}\boldsymbol{tr}\left[\boldsymbol{\Sigma}^{-1}\left(\sum_{i=1}^{n}(\boldsymbol{x}_i-\overline{\boldsymbol{x}})(\boldsymbol{x}_i-\overline{\boldsymbol{x}})'+n(\overline{\boldsymbol{x}}-\boldsymbol{\mu})(\overline{\boldsymbol{x}}-\boldsymbol{\mu})'\right)\right]}$$

Maximum Likelihood Estimator of μ

The likelihood function is

$$L(\mu,\Sigma) = (2\pi)^{-\frac{np}{2}}|\Sigma|^{-\frac{n}{2}}e^{-\frac{1}{2}\boldsymbol{tr}\left[\boldsymbol{\Sigma}^{-1}\left(\sum_{i=1}^{n}(\boldsymbol{x}_i-\bar{\boldsymbol{x}})(\boldsymbol{x}_i-\bar{\boldsymbol{x}})'+n(\bar{\boldsymbol{x}}-\boldsymbol{\mu})(\bar{\boldsymbol{x}}-\boldsymbol{\mu})'\right)\right]}$$

$$= (2\pi)^{-\frac{np}{2}}|\Sigma|^{-\frac{n}{2}}e^{-\frac{1}{2}\boldsymbol{tr}\left[\boldsymbol{\Sigma}^{-1}\left(\sum_{i=1}^{n}(\boldsymbol{x}_i-\bar{\boldsymbol{x}})(\boldsymbol{x}_i-\bar{\boldsymbol{x}})'\right)\right]}e^{-½\mathrm{tr}\left[n\boldsymbol{\Sigma}^{-1}(\bar{\boldsymbol{x}}-\boldsymbol{\mu})(\bar{\boldsymbol{x}}-\boldsymbol{\mu})'\right]}$$

$$= (2\pi)^{-\frac{np}{2}}|\Sigma|^{-\frac{n}{2}}e^{-\frac{1}{2}\boldsymbol{tr}\left[\boldsymbol{\Sigma}^{-1}\left(\sum_{i=1}^{n}(\boldsymbol{x}_i-\bar{\boldsymbol{x}})(\boldsymbol{x}_i-\bar{\boldsymbol{x}})'\right)\right]}e^{-½\left[n(\bar{\boldsymbol{x}}-\boldsymbol{\mu})\boldsymbol{\Sigma}^{-1}(\bar{\boldsymbol{x}}-\boldsymbol{\mu})'\right]}$$

Notice that the expression in terms of μ, $L(\mu,\Sigma)$ is maximized when $(\bar{\boldsymbol{x}}-\boldsymbol{\mu})\boldsymbol{\Sigma}^{-1}(\bar{\boldsymbol{x}}-\boldsymbol{\mu})'$ is minimized.

However, $\boldsymbol{\Sigma}^{-1}$ is positive definite, therefore $(\bar{\boldsymbol{x}}-\boldsymbol{\mu})\boldsymbol{\Sigma}^{-1}(\bar{\boldsymbol{x}}-\boldsymbol{\mu})' \geq 0$ and the minimum is at 0 when $\mu = \bar{x}$.

You can also prove the result with multivariate calculus:

$$L(\boldsymbol{\mu},\Sigma) = (2\pi)^{-\frac{np}{2}}|\Sigma|^{-\frac{n}{2}}e^{-\frac{1}{2}\sum_{i=1}^{n}(\boldsymbol{x}_i-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_i-\boldsymbol{\mu})}$$

$$\Rightarrow l(\boldsymbol{\mu},\Sigma) = -\frac{np}{2}\log(2\pi) - \frac{n}{2}\log|\Sigma| - \frac{1}{2}\sum_{i=1}^{n}(\boldsymbol{x}_i-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_i-\boldsymbol{\mu})$$

$$\frac{\partial}{\partial\boldsymbol{\mu}}l(\boldsymbol{\mu},\Sigma) = -\frac{1}{2}\sum_{i=1}^{n}(\boldsymbol{x}_i-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1} = 0 \text{ iff } \boldsymbol{\mu} = \bar{x}$$

Maximum Likelihood Estimator of $\boldsymbol{\Sigma}$

Recall the joint likelihood function is:

$$L(\boldsymbol{\mu}, \Sigma) = (2\pi)^{-\frac{np}{2}} |\Sigma|^{-\frac{n}{2}} e^{-\frac{1}{2}\mathrm{tr}\left[\sum_{i=1}^{n} \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_i - \boldsymbol{\mu})(\boldsymbol{x}_i - \boldsymbol{\mu})'\right]}$$

$$\Rightarrow l(\boldsymbol{\mu}, \Sigma) = -\frac{np}{2}\log(2\pi) - \frac{n}{2}\log|\Sigma| - \frac{1}{2}\mathrm{tr}\left[\sum_{i=1}^{n} \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_i - \boldsymbol{\mu})(\boldsymbol{x}_i - \boldsymbol{\mu})'\right]$$

$$= -\frac{np}{2}\log(2\pi) - \frac{n}{2}\log|\Sigma| - \frac{1}{2}\mathrm{tr}\left[\boldsymbol{\Sigma}^{-1}\boldsymbol{S}_{\boldsymbol{\mu}}\right]$$

where $\boldsymbol{S}_{\boldsymbol{\mu}} = \sum_{i=1}^{n}(\boldsymbol{x}_i - \boldsymbol{\mu})(\boldsymbol{x}_i - \boldsymbol{\mu})'$

Next, we need some multivariate calculus results from http://www2.imm.dtu.dk/pubdb/edoc/imm3274.pdf

$$\frac{\partial}{\partial X}\log|X| = (X^{-1})'$$

$$\frac{\partial}{\partial X}\mathrm{tr}(AX^{-1}B) = -(X^{-1}BAX^{-1})'$$

Maximum Likelihood Estimator of $\boldsymbol{\Sigma}$

Recall the joint loglikelihood function is:

$$l(\boldsymbol{\mu}, \Sigma) = -\frac{np}{2}\log(2\pi) - \frac{n}{2}\log|\Sigma| - \frac{1}{2}\mathrm{tr}\left[\boldsymbol{\Sigma}^{-1}\boldsymbol{S}_{\boldsymbol{\mu}}\right]$$

Therefore,

$$\frac{\partial}{\partial\Sigma}\log|\Sigma| = (\Sigma^{-1})' = \Sigma^{-1}$$

$$\frac{\partial}{\partial\Sigma}\mathrm{tr}\left(\boldsymbol{\Sigma}^{-1}\boldsymbol{S}_{\boldsymbol{\mu}}\right) = \frac{\partial}{\partial\Sigma}\mathrm{tr}\left(\boldsymbol{I}\boldsymbol{\Sigma}^{-1}\boldsymbol{S}_{\boldsymbol{\mu}}\right) = -\left(\Sigma^{-1}\boldsymbol{S}_{\boldsymbol{\mu}}\boldsymbol{I}\Sigma^{-1}\right)' = -\Sigma^{-1}\boldsymbol{S}_{\boldsymbol{\mu}}\Sigma^{-1}$$

$$\Rightarrow \frac{\partial}{\partial\Sigma}l(\boldsymbol{\mu}, \Sigma) = -\frac{n}{2}\Sigma^{-1} - \frac{1}{2}\left(-\Sigma^{-1}\boldsymbol{S}_{\boldsymbol{\mu}}\Sigma^{-1}\right)$$

$$\Rightarrow \frac{\partial}{\partial\Sigma}l(\boldsymbol{\mu}, \Sigma) = 0 \iff \Sigma^{-1}\boldsymbol{S}_{\boldsymbol{\mu}}\Sigma^{-1} = n\Sigma^{-1}$$

$$\iff \Sigma^{-1}\boldsymbol{S}_{\boldsymbol{\mu}} = n \iff \boldsymbol{S}_{\boldsymbol{\mu}} = n\Sigma$$

$$\Rightarrow \boxed{\hat{\Sigma}_{MLE} = \frac{\boldsymbol{S}_{\boldsymbol{\mu}}}{n} = \frac{1}{n}\sum_{i=1}^{n}(\boldsymbol{x}_i - \overline{\boldsymbol{x}})(\boldsymbol{x}_i - \overline{\boldsymbol{x}})'}$$

Sample Covariance Matrix

The MLE of $\boldsymbol{\Sigma}$ is:

$$\hat{\Sigma}_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(\boldsymbol{x}_i - \overline{\boldsymbol{x}})(\boldsymbol{x}_i - \overline{\boldsymbol{x}})'$$

Instead of the MLE we usually use the sample covariance matrix:

$$\boxed{\boldsymbol{S} = \frac{1}{n-1}\sum_{i=1}^{n}(\boldsymbol{x}_i - \overline{\boldsymbol{x}})(\boldsymbol{x}_i - \overline{\boldsymbol{x}})'}$$

In R it is computed with `cov`
In Python it is computed with `numpy.cov`
Both software use "n – 1" in the denominator.

Large-Sample Properties

Sampling distributions:

$$\bar{\boldsymbol{X}} \sim N_p\left(\boldsymbol{\mu}, \frac{1}{n}\boldsymbol{\Sigma}\right), (n-1)\boldsymbol{S} \sim \text{Wishart with df} = n-1$$

Also, $\bar{\boldsymbol{X}}$ and $\boldsymbol{S}$ are independent r.v.s

Multivariate Law of Large Numbers:

$$\bar{\boldsymbol{X}} \xrightarrow{P} \boldsymbol{\mu}$$

Multivariate Central Limit Theorem:

Let $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$ be iid random vectors with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. Then:

$$\sqrt{n}(\bar{\boldsymbol{X}} - \boldsymbol{\mu}) \xrightarrow{d} N_p(0, \boldsymbol{\Sigma})$$

and

$$\sqrt{n}(\bar{\boldsymbol{X}} - \boldsymbol{\mu})'\boldsymbol{S}^{-1}(\bar{\boldsymbol{X}} - \boldsymbol{\mu}) \xrightarrow{d} \chi_p^2$$

Wishart Distribution

- The Chi-square random variable with n degrees of freedom is a sum of squares of n independent standard normal variables.
- The Wishart distribution is a sum of *outer products* of normal random *vectors*.
- It is a random *matrix* which is symmetric and positive definite.
- Let $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n \sim \mathrm{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Then the distribution of the $p \times p$ random matrix $\boldsymbol{M} = \sum_{i=1}^{n} \boldsymbol{X}_i \boldsymbol{X}_i{}'$ is said to have the Wishart distribution with df = n.
- It defines the distribution of the sample covariance matrix.
- Notation: $\boldsymbol{M} \sim W_p(\boldsymbol{\Sigma}, n)$

Obtaining samples in `R`:

To generate n random matrices, distributed according to the Wishart distribution with parameters Σ and df, $W_p(\Sigma, \mathrm{m})$, where m = df.

Use Function: rWishart(n, `df`, Σ)